



LIYA NIU

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English proficiency: CET-4 563; CET-6 576



Research Interests: Climate engineering; Solar radiation modification; Marine Cloud Brightening (MCB); aerosol-cloud interactions; low clouds; satellite remote sensing; cloud radiative effects

EDUCATION

Sep 2025 - Present	Peking University	Atmospheric Physics and Atmospheric Environment	Doctor of Philosophy	Advisor: Prof. Chuanfeng Zhao
Sep 2021 - Jun 2025	China Agricultural University	Applied Meteorology	Bachelor of Science	Advisor: Prof. Chuang Zhao

SELECTED RESEARCH PROJECTS

- 1. Occurrence Characteristics and Future Projections of Compound Extreme Hot-Dry Wind Events in the North China Winter Wheat Region**
Sep 2022 - Jun 2025 | Principal Investigator
Beijing Municipal Undergraduate Research Project.
- 2. Regional Differences, Causes, and Countermeasures of Climate Warming in China**
Sep 2023 - Jan 2024 | Project Lead
Youth Science and Technology Green Innovation Competition Project.
- 3. Theory and Technology of Low-Cloud Radiative Intervention Geoengineering and Its Climate Effects**
Dec 2024 - Nov 2029 | Major Participant
National Key Research and Development Program of China.

HONORS AND AWARDS

- Outstanding Communist Party Member, School of Physics, Peking University (Mar 2026)
- Beijing Outstanding Student and Beijing Outstanding Graduate (Jun 2025)
- Hello Campus Dream-Building Scholarship (Jun 2024)
- Embry Form Scholarship (Jun 2022)
- Outstanding Student and First-Class Academic Scholarship, China Agricultural University (awarded for three consecutive years)

PUBLICATIONS AND MANUSCRIPTS

- Niu Liya, Zhao Chuanfeng*, Zhang Haotian, et al. 2026. Spatiotemporal Characteristics of Aerosols and Low Clouds from MODIS and Their Implications for Marine Cloud Brightening. *Science China Earth Sciences*. [SCI Q1 Top]
- Zhang Haotian, Zhao Chuanfeng*, Chen Annan, Niu Liya, et al. 2026. Impacts of Stratospheric Aerosol Injection Geoengineering Deployment and Termination on Extreme Climate Events in China. *Science China Earth Sciences*. [SCI Q1 Top]
- Zhao Chuanfeng*, Niu Liya, Li Jiefeng, Zhang Haotian. 2026. Marine cloud brightening: state-dependent efficacy, deployment design, and systemic risks. *Earth-Science Reviews*. [SCI Q1 Top, Review Article] (Submitted)
- Lin Ruowei+, Niu Liya+, Zhao Chuang*, et al. 2025. Climate change drives escalating hot-dry wind threat to the wheat breadbasket in China. *Nature Partner Journals: Sustainable Agriculture*. [SCI Q1 Top] (Submitted)

CONFERENCE PARTICIPATION

- Second Youth Academic Forum of the Cropping Systems Branch, Chinese Association of Agricultural Science Societies, Lanzhou, China (2023)

SELECTED COURSEWORK

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| • Atmospheric Physics | • Cloud and Precipitation Physics | • Atmospheric Remote Sensing |
| • Atmospheric Radiation and Optics | • Modern Atmospheric Observation | • Microclimatology |
| • Air Pollution Meteorology | • Atmospheric Chemistry | • Atmospheric Turbulence |
| • Dynamic Meteorology | • Geographic Information Systems | • Statistical Analysis and Applications |
| • Agricultural Meteorology | • Agricultural and Climate Statistics | • Fundamentals of Computer Simulation |